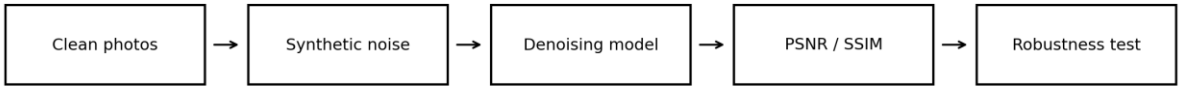
Exploratory Modeling Report

Deliverable 2 — Image Denoising



Executive Summary

Deliverable 2 addresses image denoising after photo detection and before downstream captioning. The experiment uses clean photographs, corrupts them synthetically with several realistic noise processes, and trains neural networks to reconstruct the clean target. This turns denoising into a supervised image-to-image learning problem in which the correct output is known because the noise is generated by the pipeline itself.

The dataset begins with 9,993 photographs. The quality-control stage removes 59 images whose smallest side is below 256 pixels; no unreadable files or exact duplicates were recorded in this snapshot. The resulting 9,934 images are split into 6,953 training, 1,490 validation, and 1,491 test images. The corruption generator covers five families: Gaussian, salt-and-pepper, Poisson, speckle, and JPEG compression. Training corruption remains stochastic, while validation and test corruption are deterministic for reproducible comparisons.

The architecture exploration covers a simple convolutional autoencoder, a U-Net, DnCNN, MobileNetV2 U-Net with frozen and fine-tuned encoders, ResNet50 U-Net with frozen and fine-tuned encoders, and a type-6 four-level residual U-Net (about 1.95 M parameters) that predicts the noise to subtract. Existing U-Nets can also be switched to residual prediction through the RESIDUAL option. Evaluation additionally includes eight-orientation test-time averaging (x8).

Key result	Recorded value
Best average test PSNR	U-Net ResNet50 fine-tuned — 32.0568 dB
Best average test SSIM	U-Net ResNet50 fine-tuned — 0.8890
Closest lightweight alternative	Plain U-Net — 32.0144 dB / 0.8870 SSIM
Parameter difference	ResNet50 FT has ~17.6× the parameters of plain U-Net
Strongest extreme Gaussian and salt-pepper robustness	Plain U-Net
Strongest extreme Poisson robustness	U-Net ResNet50 fine-tuned
Five-noise result	UNet_deep — 30.4973 dB / 0.8528 SSIM
Five-noise result with x8 test-time averaging	31.0264 dB / 0.8724 SSIM
x8 gain over single orientation	+0.5291 dB PSNR / +0.0196 SSIM

Across the three classic noises, the fine-tuned ResNet50 U-Net records the highest mean PSNR at 32.0568 dB, with the plain U-Net almost tied. In the broader five-noise evaluation, the type-6 residual U-Net reaches 30.4973 dB / 0.8528 SSIM from a single orientation, and 31.0264 dB / 0.8724 SSIM when eight orientation-corrected outputs are averaged. These recorded results answer complementary questions because not every architecture was retrained under exactly the same five-noise training configuration.

Report Structure

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2. Data quality, cleaning, and splitting
3. Synthetic noise generation
4. Evaluation metrics
5. Architecture paths explored
6. Training protocol and controls
7. Test-set comparison
8. Qualitative case study

9. Robustness beyond the training noise range
10. Conclusions and retained options
11. Extended five-noise protocol, residual U-Net, x8 evaluation, and GPU options

1. Business Objective and Experimental Strategy

TouNum digitizes paper archives in bulk. After Deliverable 1 identifies photographic images, Deliverable 2 cleans those photos before they are passed to the next stage of the pipeline. The practical motivation is that scanner grain, sensor noise, dust-like impulses, and other corruptions can hide useful visual content and may cause a captioning system to describe artifacts instead of the scene.

The project follows a supervised denoising strategy. A clean image is treated as the target; a noisy version is generated programmatically and used as the model input. Because the corruption is synthetic, every training pair has a known clean answer. The experimentation deliberately progresses from simple to more specialized architectures so that each added mechanism can be assessed.

Path	Core idea	Question answered
CDAE baseline	Compress and reconstruct through a bottleneck	How far can a small conventional autoencoder go?
U-Net	Add skip connections from encoder to decoder	Do direct detail-preserving shortcuts remove blur?
DnCNN	Predict the residual noise and subtract it	Is residual denoising better than reconstructing the whole image?
MobileNetV2 U-Net	Reuse pretrained early visual features	Does lightweight transfer learning improve denoising?
ResNet50 U-Net	Reuse a larger pretrained residual encoder	Does a richer pretrained encoder improve reconstruction?
Fine-tuning	Unfreeze pretrained encoders at low learning rate	Can ImageNet features be adapted to the denoising domain?
Deep residual U-Net (type 6)	Four U-Net levels; predict noise and subtract it	Can a larger residual U-Net combine skip connections with residual denoising?

2. Data Quality, Cleaning, and Splitting

2.1 Photo-only dataset

Unlike Deliverable 1, which uses five visual categories, the denoising task is trained only on the Photo folder. This is consistent with the intended pipeline: the classifier first keeps photos, then denoising operates on those retained images.

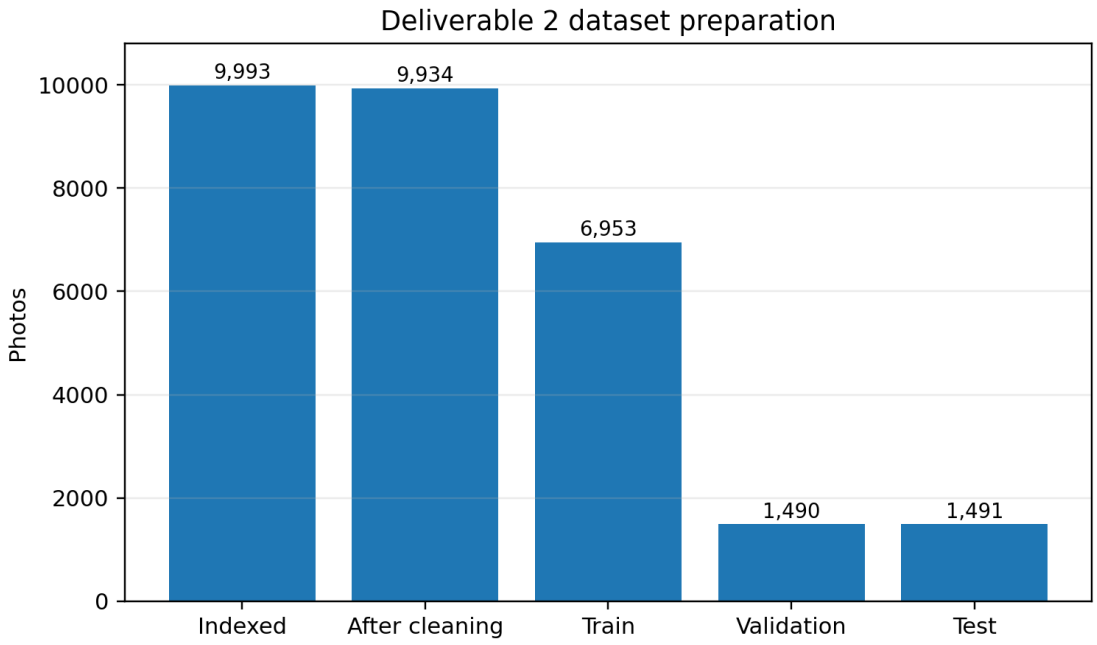


Figure 1. Photo dataset preparation and the fixed 70/15/15 split.

Quality item	Recorded value
Indexed photos	9,993
Unreadable	0
Too small (<256 px on at least one side)	59
Exact duplicates (MD5)	0
Retained after cleaning	9,934
Median size	640 × 480 px
Color modes	9,968 RGB; 25 grayscale (L)
Split	6,953 train / 1,490 validation / 1,491 test

2.2 Why the cleaning rule is stricter than Deliverable 1

For classification, a small image can still be a valid example of the class. Denoising is different: the clean image is the answer the network must reconstruct. Enlarging a photo whose smallest side is below 256 pixels would create an artificially blurred target and teach the model that blur is correct. For that reason, the denoising pipeline removes such images before the split.

- **Decoder consistency.** Images are opened with TensorFlow, matching the training decoder.
- **Duplicate prevention.** MD5 fingerprints are computed before splitting so exact copies cannot leak between train and test.
- **Fixed split.** Random seed 42 makes the same clean photographs belong to the same subsets across runs.
- **Standardized input.** Every retained image is decoded to 3-channel RGB and resized to 256×256 with antialiasing.

3. Synthetic Noise Generation

The central data-augmentation mechanism is the corruption generator itself. Each training image receives a newly sampled corruption at each epoch, with both the noise family and its strength changing. The generator covers five families by including speckle noise and JPEG compression alongside Gaussian, salt-and-pepper, and Poisson noise. This broader training distribution is intended to reduce specialization to one synthetic artifact and make the denoiser more representative of real digitization pipelines.

Noise	Training strength	Interpretation
Gaussian	σ sampled from 0.05–0.20	Fine grain added across the image; models electronic sensor/scanner noise.
Salt and pepper	1%–10% corrupted pixels	Random pixels become black or white; models dust, dead pixels, or transmission errors.
Poisson	peak 10–60	Signal-dependent counting noise; lower peak values mean stronger noise.
Speckle	σ sampled from 0.05–0.30	Multiplicative grain whose amplitude scales with the image signal; models texture-like sensor/scanner noise.
JPEG compression	quality sampled from 10–60	Block and ringing artifacts produced by lossy compression; lower quality means stronger degradation.

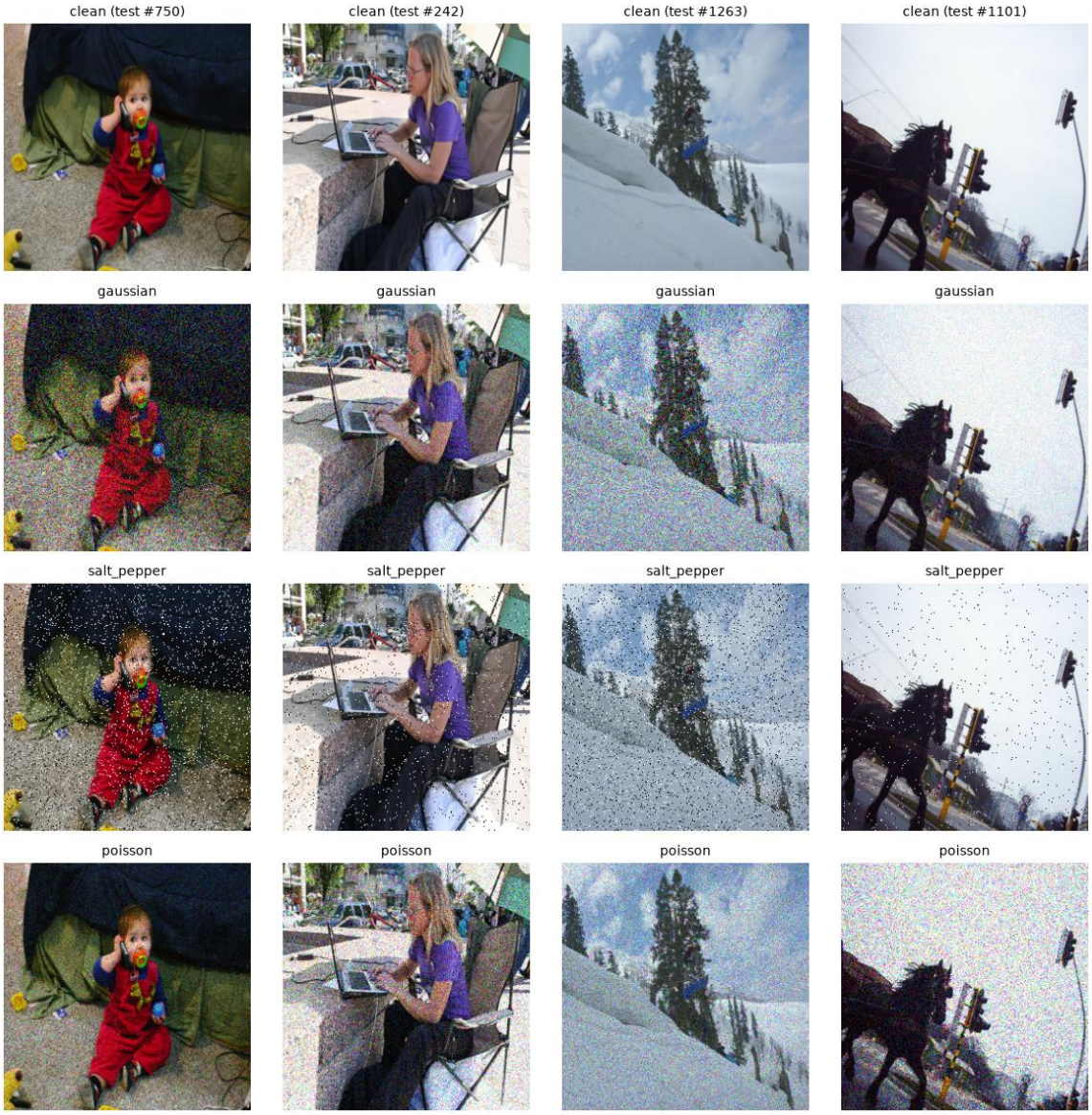


Figure 2. Control figure: clean photographs and examples of the three classic synthetic noise families.

Training noise is stochastic at every epoch. Validation noise is generated from fixed stateless seeds so the validation signal does not change between runs. The same 1,491 held-out photographs are evaluated under five deterministic corruption families. The Gaussian, salt-and-pepper, and Poisson noisy-input reference scores remain unchanged, providing a stable “3 classic” subset for comparison.

3.1 Extension from three to five noise families

The control visualization, quantitative evaluation, qualitative case study, and out-of-range robustness experiment were all extended to the same five-noise protocol. This is important methodologically: the model is no longer judged only on additive/impulse/counting noise, but also on multiplicative speckle and compression artifacts that behave differently from the original three corruptions.

The evaluation also keeps a separate “3 classic” aggregate over Gaussian, salt-and-pepper, and Poisson. The unchanged noisy-input classic baseline is 18.8773 dB PSNR and 0.3851 SSIM, which makes it possible to compare recorded experiments on the same three-noise subset without changing the reference.

4. Evaluation Metrics

The report uses both Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM), because they capture complementary properties. PSNR is directly linked to mean squared error and therefore measures average pixel fidelity. SSIM compares local luminance, contrast, edges, and structures and is generally more sensitive to perceptual distortion.

Metric	Range / unit	Interpretation
PSNR	dB; higher is better	Computed as 10 log10(1/MSE) for pixels in [0,1]. About +3 dB corresponds to roughly halving the MSE.
SSIM	0–1; higher is better	Measures structural similarity; 1 means identical local structure.

The “no denoising” input is kept as an explicit reference. On the three classic noises, its average test performance is 18.8773 dB PSNR and 0.3851 SSIM; across all five noises it reaches 21.8534 dB and 0.5185 SSIM. A useful denoiser should improve substantially on these untreated-input baselines.

5. Architecture Paths Explored

5.1 Simple convolutional autoencoder (CDAE baseline)

The baseline uses two convolution-and-pooling stages to compress a 256×256×3 image to a 64×64 latent representation, followed by a symmetric upsampling decoder. It has 75,651 parameters. The design is intentionally simple, but its bottleneck can discard high-frequency texture; the notebook therefore expects cleaner but blurrier outputs.

5.2 U-Net with skip connections

The U-Net keeps the same general encoder-decoder idea but adds skip connections from the encoder to decoder layers with matching spatial resolution. These shortcuts preserve edges and textures that would otherwise be lost in the 64×64 bottleneck. At only 112,547 parameters, it remains compact while explicitly addressing the baseline's expected blur.

5.3 DnCNN residual denoising

DnCNN changes the target of learning: instead of reconstructing the clean image directly, the network predicts the noise residual and subtracts it from the noisy input. The implementation uses 10 convolutional layers with 48 filters, BatchNormalization in the middle layers, no pooling, and 170,067 parameters. The architecture was reduced from the original paper to fit the available GPU while keeping a 21×21 receptive field.

5.4 MobileNetV2 U-Net transfer learning

The MobileNetV2 path reuses early ImageNet features at 128×128, 64×64, and 32×32 resolution, then attaches a custom U-Net decoder plus a full-resolution skip. Phase 1 freezes the encoder; phase 2 unfreezes its convolutional weights while keeping BatchNormalization layers frozen and lowers the learning rate from 10^{-3} to 10^{-5}. Both variants have 353,523 parameters.

5.5 ResNet50 U-Net transfer learning

The ResNet50 path keeps analogous feature levels from a larger pretrained encoder. The retained encoder is much richer than MobileNetV2, with a 512-channel bottleneck and substantially more pretrained weights. The full denoising model has 1,982,707 parameters. It is trained first with a frozen encoder and then fine-tuned at 10^{-5}, again keeping BatchNormalization frozen.

Evaluated variant	Parameters	Distinctive mechanism
CDAE baseline	75,651	Bottleneck reconstruction
U-Net	112,547	Skip connections
DnCNN	170,067	Residual noise prediction
U-Net MobileNetV2	353,523	Frozen lightweight pretrained encoder
U-Net MobileNetV2 FT	353,523	Fine-tuned MobileNetV2 encoder
U-Net ResNet50	1,982,707	Frozen larger pretrained encoder
U-Net ResNet50 FT	1,982,707	Fine-tuned ResNet50 encoder
U-Net deep / type 6	1,951,939	Four-level U-Net with residual noise prediction
Existing U-Nets + RESIDUAL option	unchanged architecture count	Switch target from direct clean reconstruction to predicted-noise subtraction

5.6 Type 6 — Four-level residual U-Net

The architecture exploration also introduces a deeper U-Net with four encoder/decoder levels and approximately 1.95 million parameters. Unlike the direct-reconstruction U-Net, this model predicts the corruption residual and subtracts it from the noisy input. The design therefore combines U-Net skip connections, which preserve spatial detail, with the residual-denoising idea used by DnCNN.

A new RESIDUAL option also makes this residual target available to the existing U-Net families. This isolates an important experimental question: is performance driven by the backbone architecture itself, or by asking the network to predict only the noise instead of reconstructing the full clean image?

6. Training Protocol and Controls

The shared training function uses a more adaptive optimization policy. Adam remains the optimizer, while LOSS can select "mse", "mae", or a combined L1+SSIM objective. The maximum training budget is 100 epochs. ReduceLROnPlateau halves the learning rate after 3 validation epochs without improvement, and EarlyStopping waits 10 epochs. Each training photo is also randomly flipped and/or rotated, adding geometric augmentation on top of the synthetic corruption process.

Training element	Configuration
Image size	256×256 RGB
Batch size	32
Loss	Configurable: MSE, MAE, or L1+SSIM via LOSS
Optimizer	Adam
Base learning rate	1×10^{-3}
Fine-tuning learning rate	1×10^{-5}
Maximum epochs	100
EarlyStopping	patience = 10; restore best weights
Checkpoint selection	lowest validation loss
Reproducibility	global seed = 42; deterministic val/test noise
Learning-rate schedule	ReduceLROnPlateau: divide LR by 2 after 3 stagnant validation epochs
Training augmentation	Random flips and rotations applied to each training photo
Mixed precision	Optional MIXED_PRECISION switch for faster GPU computation; disabled by default

Loss selection is now an explicit experimental control rather than a future idea. MSE emphasizes pixel-wise fidelity, MAE/L1 is less sensitive to large individual errors, and L1+SSIM combines a pixel term with a structural term. Because changing the loss also changes the optimization target, comparisons should record the selected LOSS value alongside the architecture and random seed.

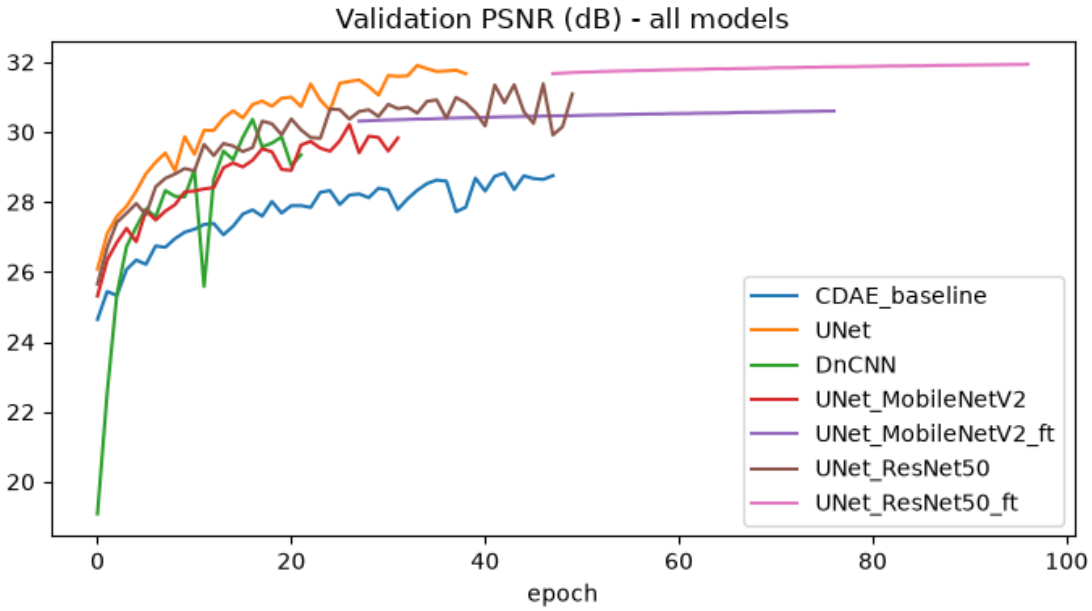


Figure 3. Validation PSNR trajectories for all evaluated models and fine-tuning phases.

The validation curves show that pretrained models begin from useful feature representations, but transfer learning is not automatically dominant. The plain U-Net reaches a similar final validation region with a much smaller architecture. Fine-tuning improves both pretrained U-Nets gradually rather than producing an abrupt jump, which is consistent with the deliberately low 10^{-5} learning rate.

6.1 Extended 100-epoch training behavior

The extended training traces show two complementary regimes. Frozen transfer models converge quickly and then plateau, while the fine-tuning phases continue making smaller improvements over the full 100-epoch budget. The type-6 U-Net shows several large validation spikes early in training but subsequently stabilizes around the 30 dB validation-PSNR region, illustrating why the more patient scheduler and stopping criteria are useful.

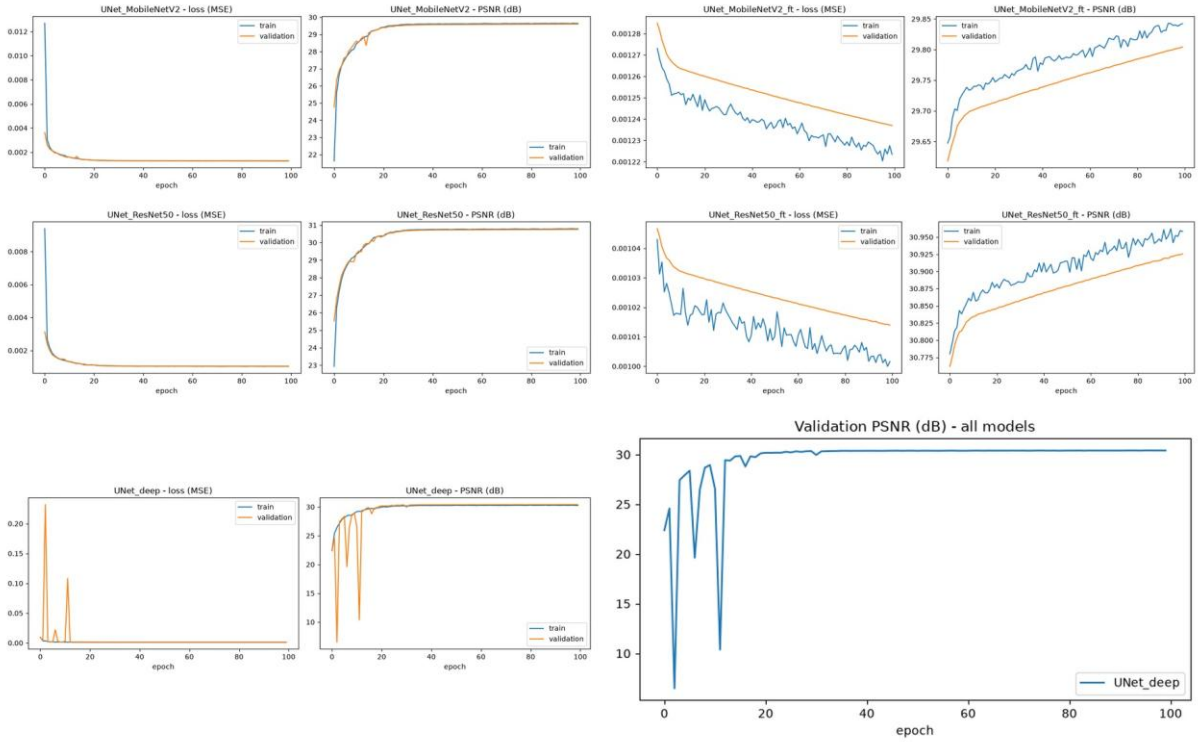


Figure 4. Training loss and PSNR traces for MobileNetV2, fine-tuned MobileNetV2, ResNet50, fine-tuned ResNet50, the type-6 U-Net, and the validation-PSNR comparison.

MIXED_PRECISION is available as an engineering option to accelerate compatible GPU workloads, but it is disabled by default so the baseline remains easy to reproduce and numerical behavior is not changed unintentionally.

7. Test-Set Results

7.1 Generalization check on the two newly introduced noise families

Previously trained denoisers can be evaluated on speckle and JPEG even when those runs did not include either corruption during training. This is a zero-shot robustness test rather than a like-for-like retraining comparison: success indicates that the learned denoising rule transfers beyond Gaussian, salt-and-pepper, and Poisson corruption.

The purpose of this subsection is therefore diagnostic. The recorded three-noise metrics provide the in-distribution reference, while performance on speckle and JPEG measures out-of-distribution generalization. A definitive training-policy gain for any architecture requires rerunning that same architecture under the full five-noise protocol.

7.2 Five-noise evaluation

The main aggregate score covers all five corruption families. For the type-6 model, the single-orientation evaluation reaches 30.4973 dB mean PSNR and 0.8528 mean SSIM. Averaging predictions over eight orientations raises these values to 31.0264 dB and 0.8724 SSIM.

Configuration	Mean PSNR (5)	Mean SSIM (5)	Mean PSNR (3 classic)	Mean SSIM (3 classic)
Noisy input	21.8534	0.5185	18.8773	0.3851
UNet_deep	30.4973	0.8528	30.3896	0.8377
UNet_deep + x8	31.0264	0.8724	31.0947	0.8642

The “3 classic” columns isolate Gaussian, salt-and-pepper, and Poisson within the same evaluation. In particular, the noisy-input baseline remains exactly 18.8773 dB / 0.3851, so comparisons on this subset use an unchanged reference.

	gaussian PSNR	gaussian SSIM	salt_pepper PSNR	salt_pepper SSIM	poisson PSNR	poisson SSIM	speckle PSNR	speckle SSIM	jpeg PSNR	jpeg SSIM	mean PSNR	mean SSIM	mean PSNR (3 classic)	mean SSIM (3 classic)	params
Noisy input (no denoising)	19.2446	0.3352	18.0573	0.4326	19.3301	0.3875	22.7577	0.5707	29.8774	0.8667	21.8534	0.5185	18.8773	0.3851	NaN
UNet_deep	28.4513	0.7756	33.9648	0.9311	28.7525	0.8064	30.9689	0.8703	30.3488	0.8804	30.4973	0.8528	30.3896	0.8377	1951939.0
UNet_deep + x8	28.9350	0.8061	35.1258	0.9528	29.2232	0.8339	31.4060	0.8872	30.4422	0.8820	31.0264	0.8724	31.0947	0.8642	1951939.0

Figure 5. Five-noise evaluation results, including the three-classic-noise subset.

7.3 Comparison on the three classic noise families

The "3 classic" subset allows recorded runs to be compared without mixing in speckle and JPEG. The comparison intentionally leaves cells blank when an architecture has not been rerun under the full five-noise protocol; this prevents a missing experiment from being misread as a measured improvement.

For the type-6 U-Net, the classic-subset score is 30.3896 dB without test-time averaging and 31.0947 dB with x8 averaging. These figures are useful for context, but they should not be interpreted as a pure architecture gain against runs trained under a different corruption distribution and optimization policy.

Model	Earlier recorded PSNR (3 classic)	Five-noise run: PSNR on 3 classic	Difference (dB)
CDAE baseline	28.939	—	—
DnCNN	30.428	—	—
Noisy input (no denoising)	18.877	18.877	0.000
U-Net	32.014	—	—
U-Net MobileNetV2	30.305	—	—
U-Net MobileNetV2 FT	30.693	—	—
U-Net ResNet50	31.481	—	—
U-Net ResNet50 FT	32.057	—	—
UNet_deep	—	30.390	—
UNet_deep + x8	—	31.095	—

Table 8. Classic-noise comparison across recorded runs. Dashes indicate that the architecture was not rerun under the five-noise training protocol.

7.4 Eight-orientation test-time averaging

The strongest five-noise model is evaluated a second time by transforming the same test photo into eight orientations, denoising each orientation, undoing the transform, and averaging the aligned outputs. This does not retrain the network; it is a test-time ensemble over geometric views of the same image.

For UNet_deep, x8 increases the five-noise mean PSNR from 30.4973 to 31.0264 dB (+0.5291 dB) and mean SSIM from 0.8528 to 0.8724 (+0.0196). On the three classic noises, the improvement is larger: 30.3896 → 31.0947 dB (+0.7051 dB) and 0.8377 → 0.8642 SSIM (+0.0265). The trade-off is roughly eight forward passes per image, so this technique is most attractive when reconstruction quality is more important than inference latency.

7.5 Reference results on the three classic noise families

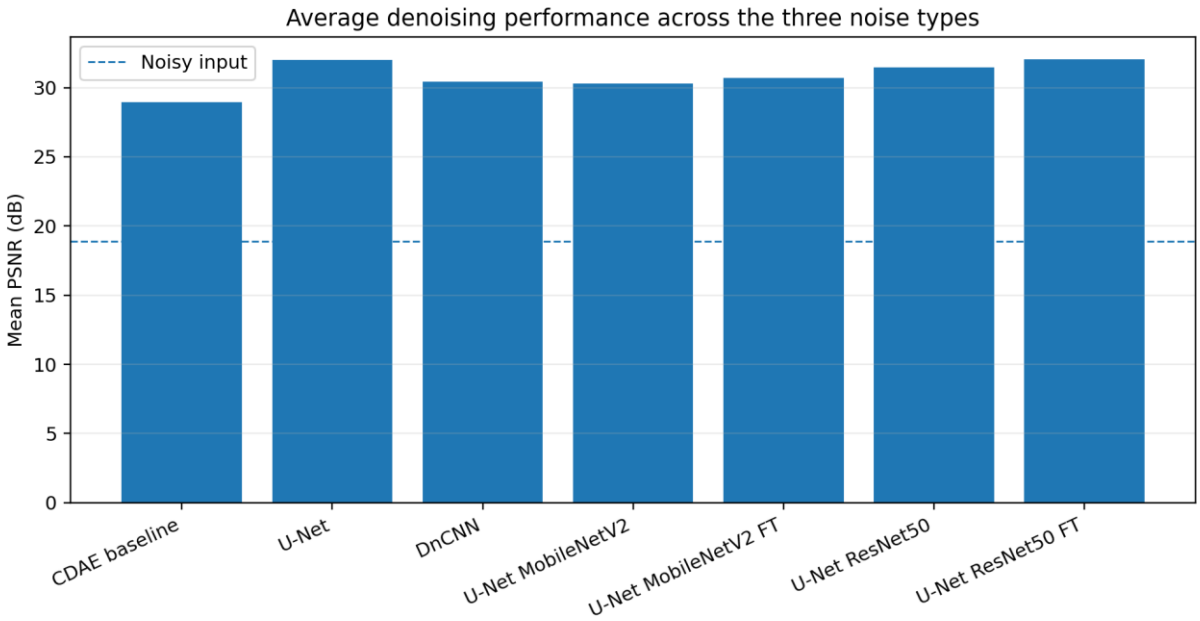


Figure 6. Mean PSNR across Gaussian, salt-and-pepper, and Poisson test sets.

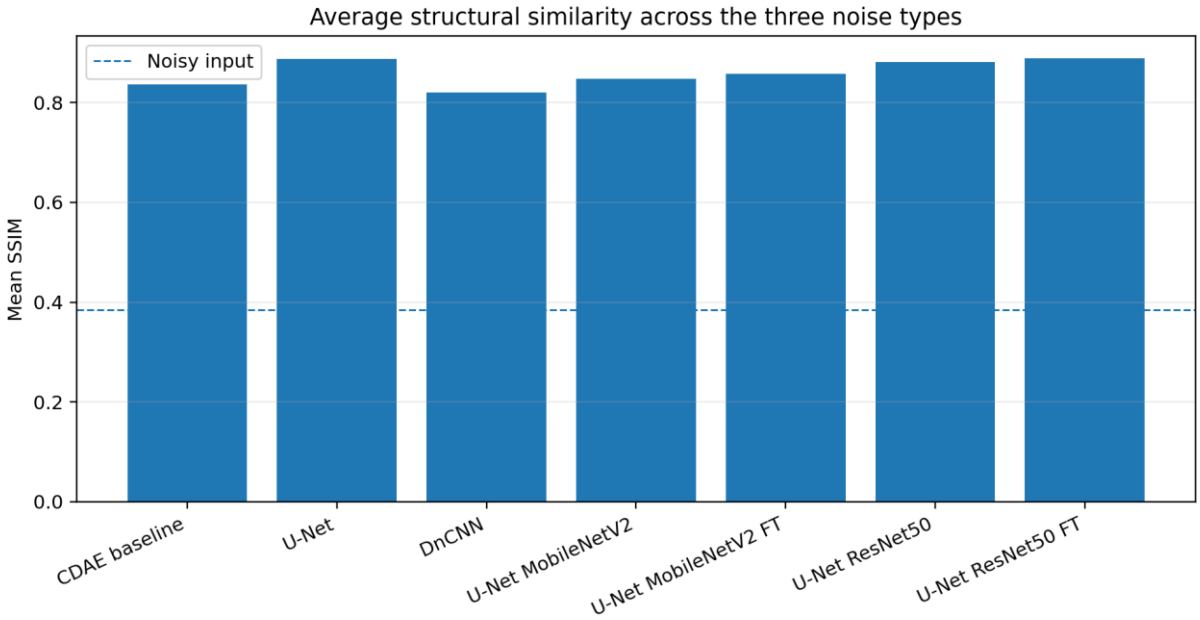


Figure 7. Mean SSIM across the same three noise families.

Model	Mean PSNR	Mean SSIM	Parameters
Noisy input	18.8773	0.3851	—
CDAE baseline	28.9394	0.8362	75,651
U-Net	32.0144	0.8870	112,547
DnCNN	30.4284	0.8209	170,067
U-Net MobileNetV2	30.3048	0.8473	353,523
U-Net MobileNetV2 FT	30.6931	0.8576	353,523
U-Net ResNet50	31.4806	0.8814	1,982,707
U-Net ResNet50 FT	32.0568	0.8890	1,982,707

On the three-noise benchmark, the fine-tuned ResNet50 U-Net is the numerical leader on the average classic metrics, but only narrowly: it is 0.0424 dB PSNR and 0.002 SSIM above the plain U-Net while using about 17.6 times as many parameters. These results are reported as the classic-noise benchmark and should be interpreted separately from the five-noise aggregate.

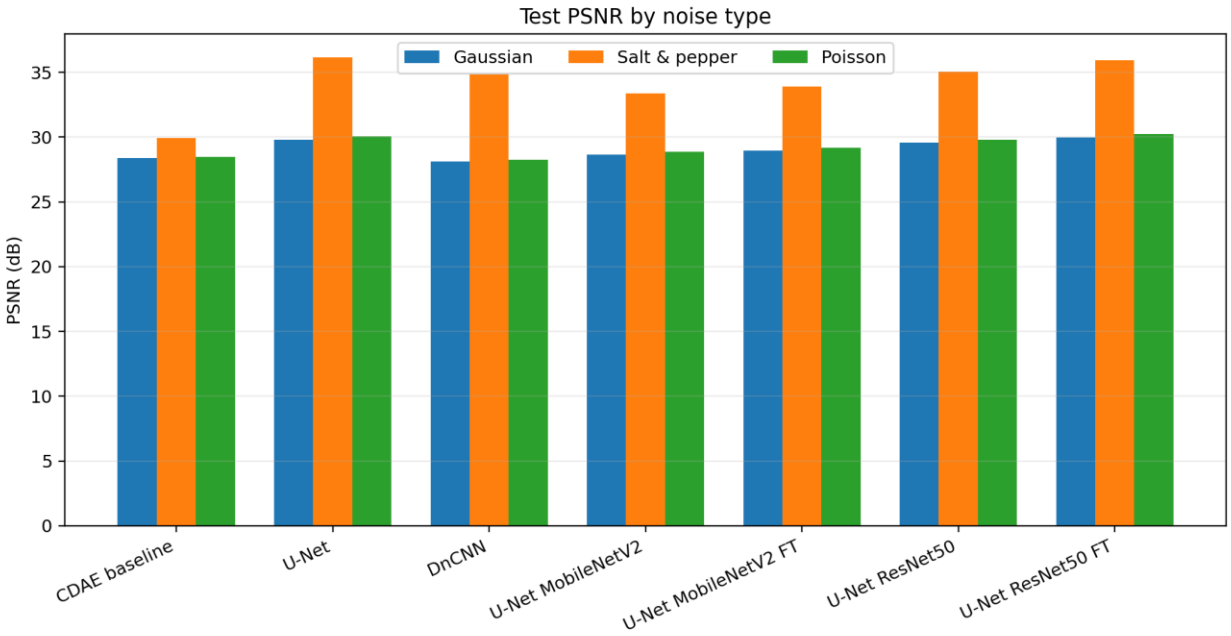


Figure 8. Test PSNR separated by corruption type.

Model	Gaussian PSNR	Salt & pepper PSNR	Poisson PSNR
CDAE baseline	28.4007	29.9394	28.4783
U-Net	29.8160	36.1564	30.0709
DnCNN	28.1436	34.8832	28.2585
U-Net MobileNetV2	28.6546	33.3761	28.8838
U-Net MobileNetV2 FT	28.9599	33.9157	29.2039
U-Net ResNet50	29.5831	35.0692	29.7896
U-Net ResNet50 FT	29.9938	35.9534	30.2233

The ranking is not identical for every corruption. Fine-tuned ResNet50 is best on Gaussian and Poisson noise, while the plain U-Net is best on salt-and-pepper noise (36.1564 dB). This is a useful warning against selecting a denoiser from a single aggregate score when the deployment noise distribution is not known in advance.

7.6 Transfer-learning and fine-tuning observations

Fine-tuning consistently helps the pretrained encoders. MobileNetV2 improves from 30.3048 to 30.6931 dB mean PSNR (+0.3883 dB) and from 0.8473 to 0.8576 SSIM. ResNet50 improves from 31.4806 to 32.0568 dB (+0.5762 dB) and from 0.8814 to 0.8890 SSIM. However, pretrained features are not automatically superior to a task-specific U-Net trained from scratch: the frozen MobileNetV2 variant is clearly below the plain U-Net, and even the fine-tuned MobileNetV2 remains below it.

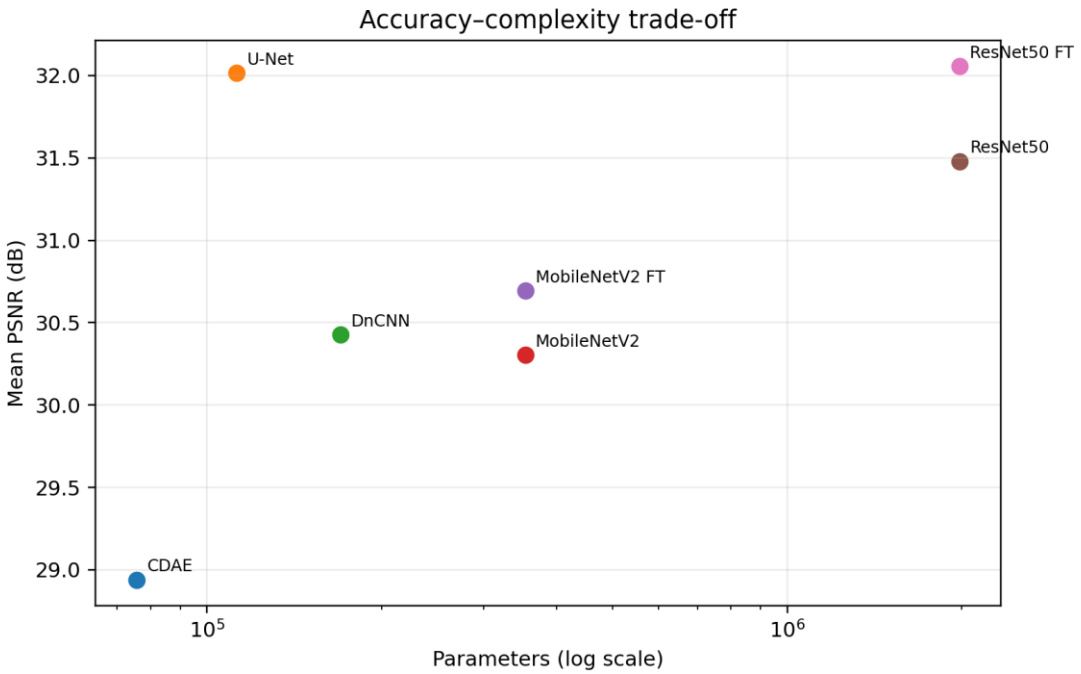


Figure 9. Mean PSNR versus parameter count, highlighting the strong efficiency of the plain U-Net.

8. Qualitative Case Study

The qualitative analysis includes both a three-noise case study and a five-noise view. The five-noise figure focuses on the type-6 U-Net and shows the clean target, noisy input, and reconstruction for Gaussian, salt-and-pepper, Poisson, speckle, and JPEG corruption, with PSNR and SSIM reported on each panel.

8.1 Three-noise case study

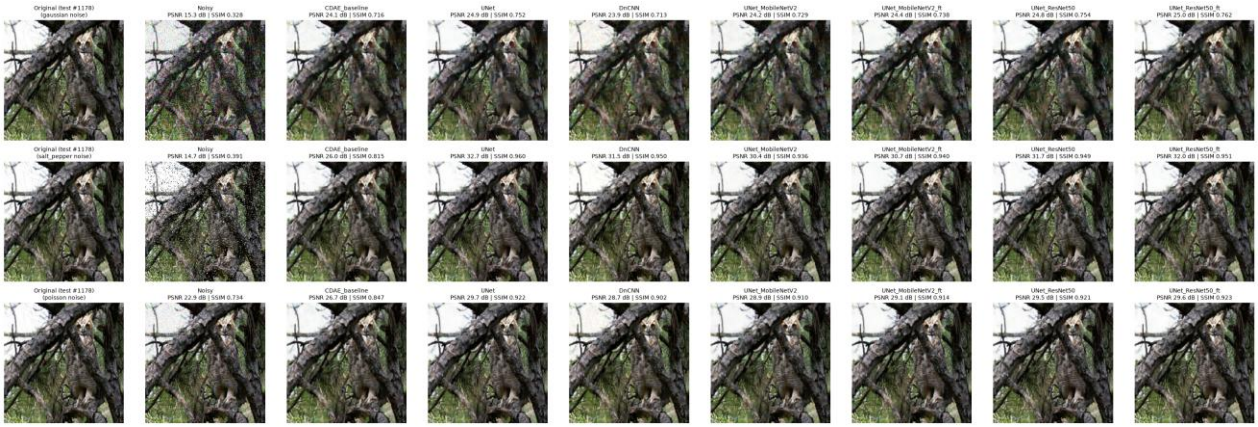


Figure 10. Case study under Gaussian, salt-and-pepper, and Poisson corruption.

8.2 Five-noise case study

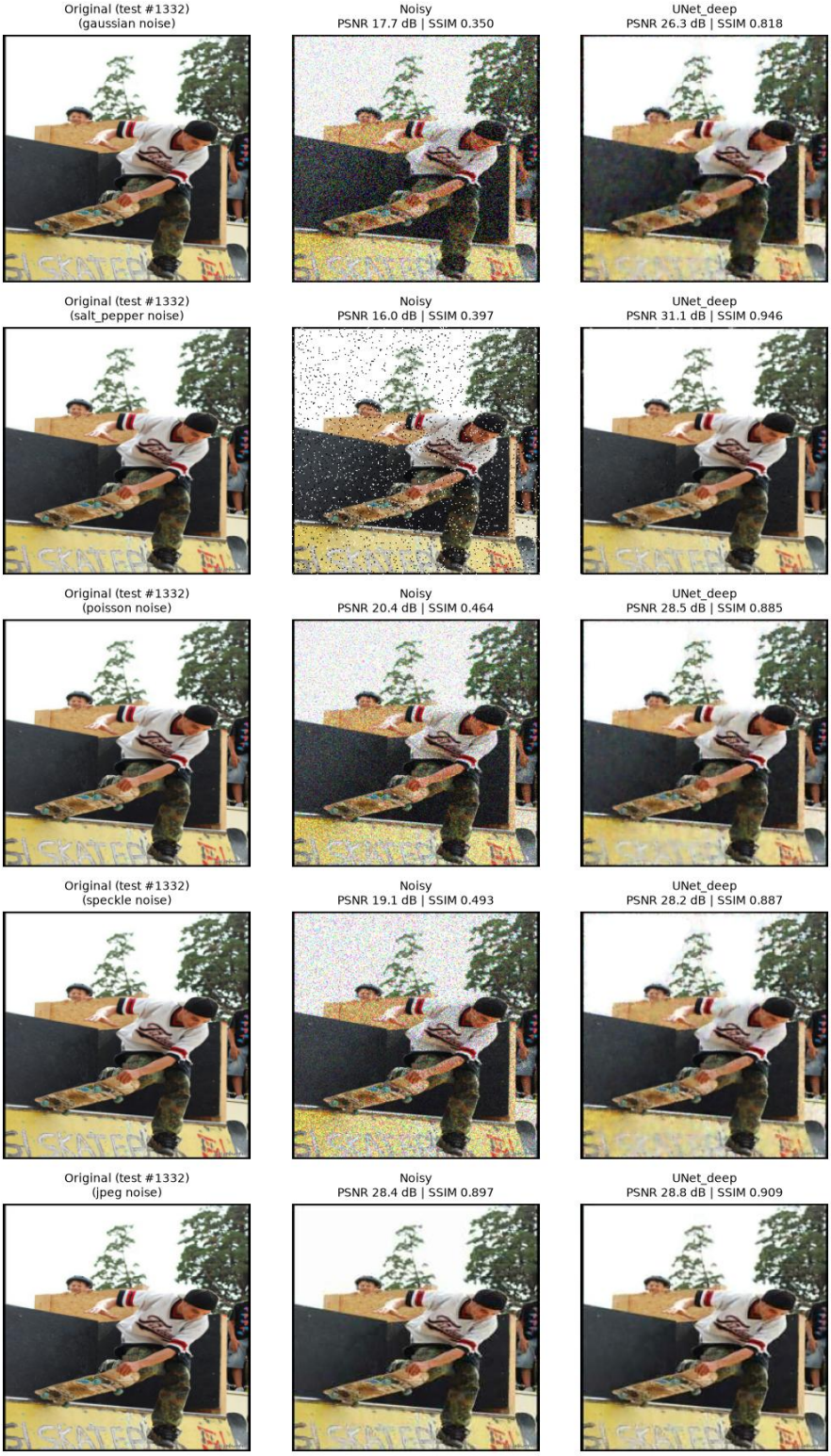


Figure 11. Five-noise qualitative case study for UNet_deep: Gaussian, salt-and-pepper, Poisson, speckle, and JPEG.

The five-noise case study shows that one aggregate number can hide very different behaviors. Salt-and-pepper impulses are removed very effectively, Gaussian and speckle noise are strongly reduced but can leave a smoother appearance, and JPEG is already relatively high-quality before denoising so the

numerical gain is modest. Poisson remains the most sensitive corruption when its strength becomes extreme.

9. Robustness Beyond the Training Noise Range

The robustness experiment evaluates all five corruption families at progressively stronger levels, including strengths beyond the ranges used during training. Nothing is retrained. This tests whether the type-6 U-Net has learned a general denoising rule or only the corruption regime seen during optimization.

Noise	Inside training range	Outside training range
Gaussian σ	0.10, 0.20	0.30, 0.40
Salt & pepper	5%, 10%	20%, 30%
Poisson peak	30, 10	5, 2 (lower = noisier)
Speckle σ	0.10, 0.30	0.45, 0.60
JPEG quality	40, 15	8, 5 (lower = stronger compression)

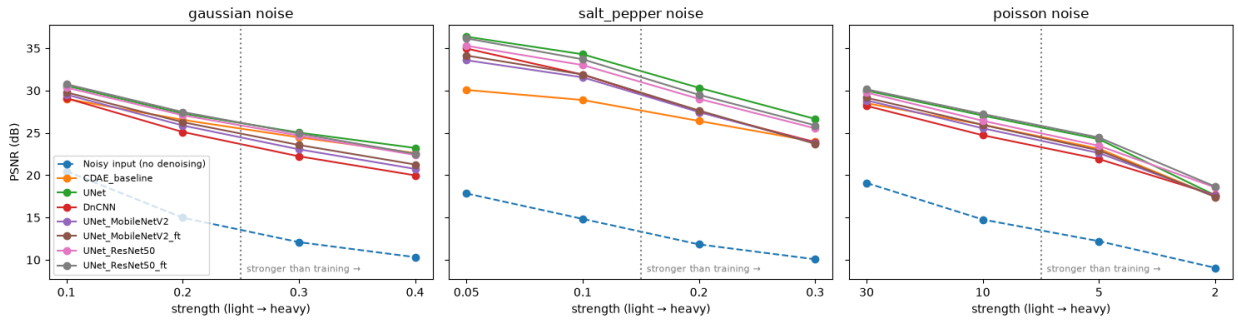


Figure 12. PSNR as corruption strength increases. The dotted boundary separates strengths seen during training from stronger out-of-range noise.

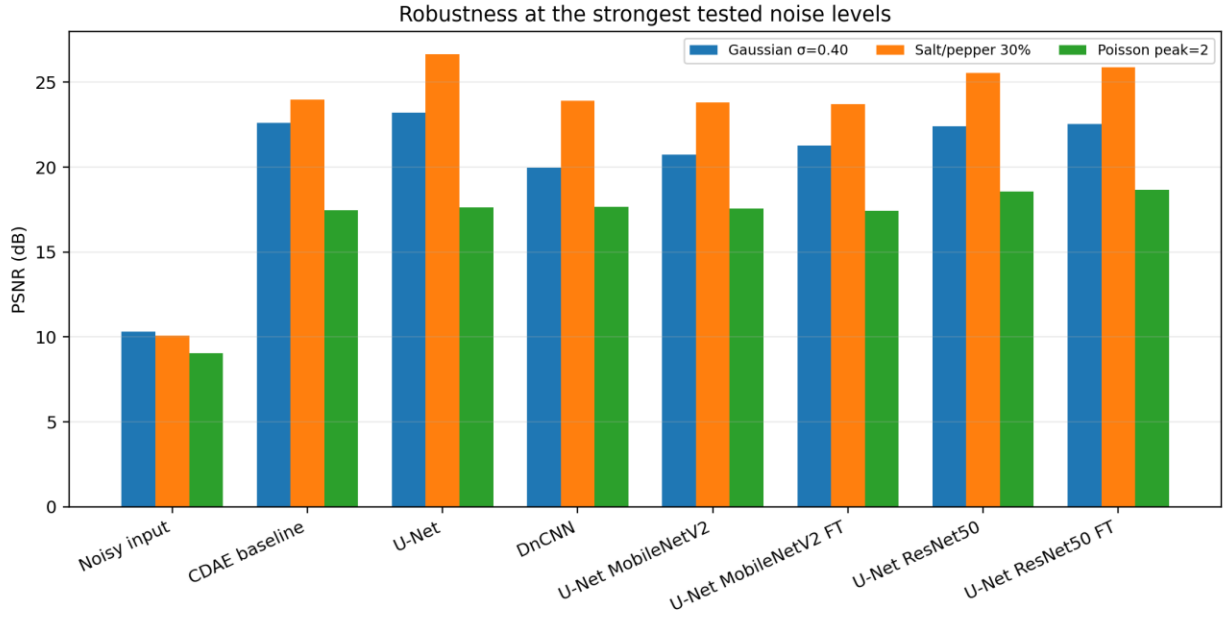


Figure 13. PSNR at the strongest tested corruption level for each noise family.

The type-6 U-Net remains useful beyond the training range for four of the five corruption families. At the strongest tested Gaussian level (σ=0.40), PSNR rises from 10.319 to 16.510 dB (+6.191 dB); at 30% salt-and-pepper it rises from 10.068 to 23.130 dB (+13.062 dB); at speckle σ=0.60 it rises from 12.609 to 20.181 dB (+7.572 dB); and at JPEG quality 5 the gain is small but positive (23.528 → 23.857 dB). Extreme Poisson noise is the important failure case: at peak=2, PSNR falls from 9.055 to 7.686 dB even though SSIM improves from 0.088 to 0.349. The qualitative image shows strong saturation/artifacts, marking a clear extrapolation limit.



Figure 14. Qualitative outputs for one random test photograph at the heaviest tested noise level for each corruption family.

noise	gaussian				salt_pepper				poisson				speckle				jpeg			
level	0.10	0.20	0.30	0.40	0.05	0.10	0.20	0.30	30.00	10.00	5.00	2.00	0.10	0.30	0.45	0.60	40.00	15.00	8.00	5.00
model																				
Noisy input (no denoising)	20.477	14.984	12.092	10.319	17.845	14.837	11.829	10.068	19.087	14.765	12.205	9.055	26.539	17.536	14.464	12.609	30.569	27.628	25.408	23.528
UNet_deep	29.381	25.393	21.937	16.510	34.186	31.359	26.849	23.130	28.669	25.126	21.894	7.686	33.267	27.604	24.745	20.181	31.046	28.121	25.829	23.857

noise	gaussian				salt_pepper				poisson				speckle				jpeg			
level	0.10	0.20	0.30	0.40	0.05	0.10	0.20	0.30	30.00	10.00	5.00	2.00	0.10	0.30	0.45	0.60	40.00	15.00	8.00	5.00
model																				
Noisy input (no denoising)	0.374	0.192	0.119	0.083	0.417	0.248	0.133	0.087	0.377	0.227	0.154	0.088	0.694	0.400	0.296	0.235	0.885	0.807	0.732	0.663
UNet_deep	0.819	0.649	0.488	0.372	0.939	0.883	0.742	0.583	0.809	0.648	0.494	0.349	0.923	0.779	0.676	0.616	0.897	0.826	0.753	0.682

Figure 15. PSNR and SSIM robustness tables for five corruption families and progressively stronger severity levels.

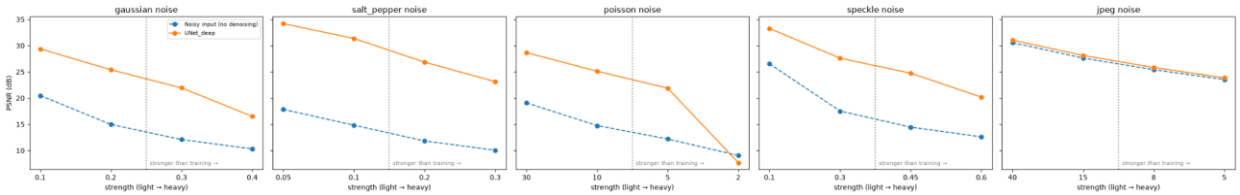


Figure 16. Five-noise robustness curves. The vertical marker separates the in-training range from stronger extrapolation tests.



Figure 17. Qualitative outputs at the strongest tested Gaussian, salt-and-pepper, Poisson, speckle, and JPEG settings.

10. Conclusions and Retained Options

The final experimental scope covers five noise families and includes a larger residual U-Net. The type-6 architecture has about 1.95 million parameters, predicts the noise residual, and can be evaluated either once or with x8 orientation averaging. Its single-orientation five-noise result is 30.4973 dB / 0.8528 SSIM; x8 raises this to 31.0264 dB / 0.8724 SSIM.

Residual learning is now explored in two ways: DnCNN predicts noise directly at full resolution, while the new type-6 U-Net combines residual prediction with a four-level encoder-decoder and skip connections. The RESIDUAL switch also allows the existing U-Nets to be tested under the same target formulation, making residual-vs-direct reconstruction an explicit experimental factor rather than an architecture-specific property.

The recorded results answer complementary questions. The strongest classic-three-noise score is the fine-tuned ResNet50 U-Net at 32.0568 dB / 0.8890 SSIM, while UNet_deep + x8 is the strongest configuration shown across all five noises. Its 31.0947 dB score on the classic subset should not be treated as a pure regression or improvement against models trained under a different noise distribution and optimization schedule.

10.1 Selection guidance

Priority	Recommended recorded option	Reason
Maximum average test metrics	U-Net ResNet50 fine-tuned	Best mean PSNR and SSIM; best Gaussian/Poisson in-range scores.
Compact deployment / strong efficiency	Plain U-Net	Near-identical average quality with ~17.6× fewer parameters.
Extreme Gaussian / impulse robustness	Plain U-Net	Highest PSNR at σ=0.40 Gaussian and 30% salt-and-pepper.
Extreme Poisson robustness	U-Net ResNet50 fine-tuned	Highest PSNR at Poisson peak=2.
Five-noise quality	UNet_deep + x8	Best supplied five-noise aggregate: 31.0264 dB / 0.8724 SSIM.
Lower-latency five-noise option	UNet_deep single orientation	One forward pass: 30.4973 dB / 0.8528 SSIM.

Fine-tuned ResNet50 remains the metric leader on the classic three-noise benchmark. Across the five-noise evaluation, UNet_deep + x8 is the strongest evaluated configuration shown in the supplied results, while single-orientation UNet_deep is the lower-latency alternative. The final deployment choice should therefore state whether the priority is maximum quality on the classic three-noise benchmark or broader robustness across five noise families.